

SnowPro Advanced: Architect — Prep Guide

PJ's Academy · The expert-level Snowflake credential.

The Architect exam (ARA-C01) proves you can **design end-to-end Snowflake platforms** — accounts, security, performance, cost governance, and data sharing at enterprise scale.

Prerequisite: SnowPro Core certified.

Exam At A Glance

Item	Detail
Exam code	ARA-C01
Questions	~65
Duration	115 minutes
Cost	\$375 USD
Prerequisite	SnowPro Core

Domain Weightings

#	Domain	Weight
1	Accounts & Security	28%
2	Performance & Scalability	20%
3	Data Engineering	20%
4	Storage, Data Protection & Sharing	16%
5	Snowflake Platform	16%

Domain 1 — Accounts & Security (28%)

- **Multi-account architecture** — organizations, ORGADMIN, account provisioning across regions/clouds.
- **RBAC at scale** — functional vs access role separation, role explosion avoidance, role inheritance design.
- **Advanced security** — Tri-Secret Secure, customer-managed keys, network policies, private connectivity (PrivateLink/Private Service Connect).
- **Data governance** — tags, tag-based masking, classification, access history, `ACCESS_HISTORY` for auditing.
- **Authentication** — SSO/SAML, SCIM provisioning, key-pair, OAuth flows.

Domain 2 — Performance & Scalability (20%)

- Warehouse sizing & multi-cluster scaling policies for mixed workloads.
- Workload isolation — dedicated warehouses per team/workload.
- Clustering, search optimization, and materialized view trade-offs at scale.
- Query acceleration service.
- Concurrency and queuing design.

Domain 3 — Data Engineering (20%)

- Ingestion architecture — Snowpipe vs Snowpipe Streaming vs Kafka connector.
- Pipeline design — Streams+Tasks vs Dynamic Tables; when to use each.
- External tables, Iceberg, and data lake integration.
- Snowpark for large-scale transformation.

Domain 4 — Storage, Protection & Sharing (16%)

- Time Travel/Fail-safe cost modelling.
- Replication, failover groups, business continuity / DR design (RPO/RTO).
- Secure Data Sharing architecture — shares, reader accounts, Data Exchange, Marketplace listings.
- Cross-cloud / cross-region sharing.

Domain 5 — Snowflake Platform (16%)

- Cost governance — resource monitors, org-level budgets, chargeback via ACCOUNT_USAGE.
- Native Apps framework & Streamlit in Snowflake.
- Cortex (AI/ML functions) at a high level.
- Partner ecosystem integration (dbt, Fivetran, Tableau).

High-Yield Architect Facts

- **Failover groups** replicate account objects for DR; **replication groups** replicate databases.
- **PrivateLink** provides private connectivity — no public internet.
- **ACCESS_HISTORY** shows who accessed what (column-level lineage) — key for governance.
- **Reader accounts** are provider-managed & billed to the provider — for non-Snowflake consumers.
- **Query Acceleration Service** offloads scan-heavy portions of queries.
- Design for **workload isolation**: separate warehouses prevent one team starving another.

Practice Questions (8)

- Q1.** Which provides private, non-internet connectivity to Snowflake? A) Network policy B) PrivateLink C) OAuth D) SSO
- Q2.** Which replicates account-level objects for disaster recovery? A) Replication group B) Failover group C) Share D) Clone
- Q3.** To audit column-level data access you query: A) QUERY_HISTORY B) ACCESS_HISTORY C) COPY_HISTORY D) LOGIN_HISTORY
- Q4.** Reader accounts are billed to: A) The consumer B) The data provider C) Snowflake D) Split
- Q5.** Best practice to prevent one team's queries starving another: A) One big warehouse B) Workload isolation via separate warehouses C) Bigger warehouse D) More Time Travel

- Q6.** Tag-based masking allows: A) Masking by IP B) Applying masking policies via object tags at scale C) Row filtering D) Encryption
- Q7.** For non-Snowflake consumers of shared data, use: A) Direct share B) Reader account C) Marketplace only D) CSV export
- Q8.** Which offloads scan-heavy query portions to extra serverless compute? A) Materialized view B) Query Acceleration Service C) Result cache D) Clustering

✓ **Answers**

1-B · 2-B · 3-B · 4-B · 5-B · 6-B · 7-B · 8-B



4-Week Plan

- **Week 1:** Accounts & Security (RBAC design, governance).
 - **Week 2:** Performance & Data Engineering (pipelines, isolation).
 - **Week 3:** Storage/Sharing/DR + Platform features.
 - **Week 4:** Full review + scenario practice, book the exam.
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